

Decision Status for No-Refolding and Lawful Churn Terms in HSF

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Abstract

This note converts the current no-refolding discussion into a decision-status document. Its purpose is not to prove the final HSF churn law, but to force a clean split between: (1) what is now treated as a framework commitment, (2) what is a plausible but not yet final interpretation, and (3) what remains unresolved. The note also records which churn-rule terms are mandatory, which current proxies are acceptable as temporary stand-ins, and which proxy families still need derivation before they should be trusted as part of a lawful implementation.

1 Purpose

The immediate problem is not lack of ideas. The problem is drift. Without a decision document, no-refolding can slide between several meanings: factorization stability, interface commitment, fermionic bookkeeping, graph persistence, or general “don’t rewrite working structure” intuitions.

This note is meant to stop that drift.

The goal is to answer, as cleanly as possible:

1. What do we currently mean by no-refolding?
2. What are we willing to treat as framework commitments right now?
3. Which churn-rule terms are lawful to implement now?
4. Which terms still need derivation before they deserve implementation?

2 Executive decision summary

The current best working stance is:

No-refolding is provisionally treated as a general dynamical protection principle for committed collective constraint-relief structures.

Corollaries:

1. A mesoscopic object that is efficiently carrying local constraint stress is not treated as arbitrary support arrangement.
2. Destroying or rewriting such a structure is not free.
3. Lawful re-expression remains possible, provided the same relief function is preserved elsewhere or maintained in another form.

4. Jordan–Wigner string structure is treated, for now, as a *candidate fermionic realization* of no-refolding, not yet as the whole general constraint.

This note therefore commits, for now, to the following hierarchy:

1. **General principle first:** protection of committed collective relief structure.
2. **Fermionic specialization second:** JW-like bookkeeping may be one concrete case.
3. **Strict topological interpretation later, if earned:** not assumed now.

3 Decision table for no-refolding

Question	Status	Current decision / working answer
Protected object	Required	The best current answer is: a <i>committed collective constraint-relief structure</i> . This may include support patches, links, ordering/bookkeeping data, and shell/core organization, but is not reduced to any one of them yet.
Kinematic or dynamical?	Required	Treat no-refolding as <i>dynamical</i> for now. It is a cost/barrier/protection principle, not yet a hard absolute ban.
General or fermionic-specific?	Required	Treat no-refolding as <i>general first</i> . Fermionic/JW structure is a candidate specialization, not yet the universal definition.
Lawful re-expression allowed?	Required	Yes. A move may demote, weaken, or reorganize support if the same collective relief function is preserved or re-expressed elsewhere sufficiently well.
Arbitrary erasure / rewriting allowed?	Required	No. A move that dismantles an effective collective relief structure without preserving its function elsewhere should incur a substantial cost and count as refolding.
JW strings central?	Plausible	Yes, likely central in fermionic sectors. But they are not yet adopted as the sole general meaning of no-refolding.
Topological protection?	Open	Maybe in some regimes, but not yet claimed. No actual invariant or protected sector has been identified.
Unique relief required?	Open	Probably not literal uniqueness. Replace “uniquely relieves tension” with “strongly preferred local solution” or “locally optimal relief structure.”
Primary operational witness	Open	Not fixed yet. Most likely a combination of: organizer fidelity loss, local MI loss, interface-role disruption, and loss of collective relief quality under a move.

4 Working formal statement

The current working statement to carry forward is:

No-refolding (working decision). If a local mesoscopic support pattern functions as a stable or metastable collective relief of substrate constraint stress, then its organization is physically committed. Subsequent dynamics may strengthen, weaken, or lawfully re-express that organization, but may not arbitrarily erase or refactor it without incurring a corresponding dynamical cost associated with loss of relief function and disruption of committed bookkeeping structure.

This statement is now treated as the current *framework-level commitment*, subject to later refinement.

5 Lawful versus unlawful moves

For future derivation and code work, the distinction should be:

5.1 Lawful re-expression

A move counts as lawful re-expression if it:

1. preserves the collective relief function of the object,
2. preserves enough relational/bookkeeping structure for physical continuity,
3. does not simply delete the organization without replacement.

5.2 Unlawful refolding

A move counts as unlawful refolding if it:

1. treats a committed collective relief structure as arbitrary,
2. destroys its functioning organization without preserving that function elsewhere,
3. removes bookkeeping/ordering/interface structure as though it were mere relabeling.

6 Mandatory churn-rule term classes

The churn rule should not yet be considered fully derived. However, several *term classes* are now mandatory if the rule is to remain faithful to HSF rather than collapse back into ad hoc scripting.

Mandatory classes

1. **Expression / adequacy term.**
The move rule must reward making relational structure explicit when doing so improves constrained expression.
2. **Finite-bandwidth burden term.**
The rule must penalize broad, overstuffed, too-many-channel support/interface structure.

3. **No-refolding burden term.**

The rule must penalize moves that destroy or incoherently rewrite committed collective relief structure.

4. **No-forgetting penalty term.**

The rule must penalize moves that destroy relational record instead of lawfully re-expressing it.

5. **No-signaling / spread burden term.**

The rule must penalize excessive spread or uncontrolled delocalization of influence/support.

6. **Counterfactual branch comparison.**

Moves must be judged by comparing a baseline branch with a moved branch, not by separate hand-written event laws.

7. **One local move principle for all moves.**

Birth, weakening, demotion, and similar updates must all be judged by the same local principle.

7 Derivation matrix: current proxies and upgrade targets

Mandatory class	term	Current acceptable temporary proxies	Upgrade target / what still needs derivation
Expression adequacy	/	Temporary proxy Local edge MI; parent-interface correlator; local retained MI under branch comparison. These are strong enough to use now.	Needs derivation A cleaner expression functional centered on faithful local relational expression, with shell/core structure demoted to secondary support rather than primary foundation.
Finite-bandwidth burden		Temporary proxy Influence-map rank/effective-rank family; mean active-channel count; thin-link activity measures.	Needs derivation A more explicit derivation of which influence-map functional is the correct bandwidth burden, and how to normalize it across scales.
No-refolding burden		Temporary proxy At most, crude interface-role overlap loss or shell/core disruption under a move. These may be used only as temporary stand-ins.	Needs derivation A proper witness of destruction of committed collective relief structure. In fermionic sectors this may require JW/string-preservation diagnostics; in the general case it likely requires function-preservation plus interface/bookkeeping-overlap measures.
No-forgetting penalty		Temporary proxy Organizer fidelity under branch comparison; local MI loss; recoverable relational-loss proxies.	Needs derivation A sharper distinction between lawful re-expression and true forgetting, ideally tied to recoverability rather than just state overlap.

Mandatory class	term	Current acceptable temporary proxies	Upgrade target / what still needs derivation
No-signaling / spread burden		Temporary proxy Local spread from the dominant core; influence-map channel spread/effective-rank spread.	Needs derivation A robust spread measure that does not depend too heavily on fragile spectral embeddings and that tracks bounded influence more directly.
Link / interface quality		Temporary proxy Endpoint influence-map activity; link purity; thin active-channel quality.	Needs derivation A decomposition into distinct lawful terms: activity, selectivity, coherence, and commitment, rather than one generic composite link score.
Organizer / structural coherence		Temporary proxy Shell size, triangle count, component coherence, degree support. Acceptable only as mesoscale support proxies, not as final primitives.	Needs derivation A derivation of which structural features are physically meaningful collective witnesses versus merely convenient graph descriptors.

8 What is lawful to implement now?

For the next lawful implementation pass, the following should be allowed:

1. Use **MI and correlator terms as temporary expression proxies**.
2. Use **influence-map rank/effective-rank family as temporary bandwidth proxies**.
3. Use **organizer fidelity and local MI loss as temporary no-forgetting proxies**.
4. Use a clearly labeled temporary no-refolding proxy only if it is explicitly acknowledged as provisional.
5. **Do not present shell size, core score, or graph persistence alone as the meaning of no-refolding.**
6. **Do not present any chosen weights as derived unless they actually are.**

In short:

The next lawful code may use temporary proxies, but only if they are openly labeled temporary, and only if the no-refolding term is not falsely presented as solved.

9 What must not be claimed yet

The following claims remain off limits for now:

1. That no-refolding has been fully derived.
2. That JW strings have already been shown to be the whole general content of no-refolding.

3. That the exact weights of the churn rule are first-principles quantities.
4. That shell/core graph measures are themselves the physical meaning of commitment.
5. That topological protection has already been established.

10 Immediate next derivation task

The next derivation task should be:

Derive or sharply constrain a minimal operational witness for no-refolding.

That witness should answer:

1. What exactly counts as disruption of committed collective relief?
2. What exactly counts as lawful preservation or re-expression of that relief?
3. In fermionic sectors, how should JW/string preservation enter the witness?

Until that is done, no-refolding remains a real framework commitment but an incomplete operational term.

11 Final status statement

The framework commitment we now adopt is:

No-refolding is provisionally understood as dynamical protection of committed collective constraint-relief structure.

The implementation commitment we now adopt is:

Only those churn-rule terms that are clearly mandatory from HSF, or clearly marked as temporary proxies for mandatory terms, should be implemented in the next lawful script.

The unresolved derivation commitment we now adopt is:

The no-refolding witness is not yet derived. It is the highest-priority upgrade target.

One-sentence bottom line

The current official position is that no-refolding protects committed collective relief structure in general, while its exact operational witness—especially in relation to JW-string or other bookkeeping structure—remains to be derived.